

Detailed Description (max. 6 pages)

Objectives: Generate novel theoretically-sound and analytics-based equity indicators to assess the performance of public transit systems in Canada. Investigate their reliability by working with planners and transit users in areas with equity gaps. Create open-source tools, including dashboards, to make these indicators available to stakeholders. Assess the effectiveness and accessibility of the tools in consultation with stakeholders and incorporate feedback in newer releases of the tools.

Rationale: Traditional public transport performance evaluations usually rely on operational efficiency (1–3), reliability (4), safety (5), and objective comfort, but often leave aside equity considerations related to both the spatial and temporal coverage of transit service (6,7). Consequently, they often fail to meet the needs of the most disadvantaged residents (8). The transit planning process primary focus is on service cost (9–11), operational features (12–14), and emissions (12,15,16). Furthermore, current standards (e.g., Transit Capacity and Quality of Service Manual (TCQSM)) and manuals (e.g., Federal Transit Administration (FTA)) completely overlook equity dimensions, yet equity-deserving groups are those who rely most on public transit (17–20). When service is insufficient (21,22), they face the greatest risk of social isolation due to transport-related factors, preventing them from participating in essential activities (8,23,24). This group, characterized as captive-users, faces greater limitations caused by disparities in transit level of service (LoS), both spatially and temporally.

Approaches: The project aims to produce actionable knowledge to support the development of more equitable transit in Canada. At one level, we will produce equity indicators grounded in sound theories of justice. We consider this aspect of the project the one of greatest academic interest. Further, to ensure that the indicators are of practical relevance, they will be developed in partnership with stakeholders. A key aspect of procedural justice is ensuring that all stakeholders participate in the planning process and have access to the information they need for decision-making. For this reason, we: 1) adopt a codesign framework to design equity indicators in partnership with stakeholders in Hamilton, Ontario; and 2) propose to develop open-source tools to mobilize this knowledge. General Transit Feeds Specifications (GTFS) are the standard for evaluation and planning of transit operations, and will be used to implement an equity analytical module to complement the usual cost-based evaluation of transit systems. To facilitate the exploration of indicators and the evaluation and planning with the equity evaluation module, we will develop open-source tools (EquiTransit) that will embed all elements in an accessible interface (a dashboard). The tool will be based on open data and will be able to use data from different regions. The aim of the tools is to allow for the examination of socioeconomic and spatial-temporal inequalities in public transportation services, based on geo-spatial technologies.

Anticipated Benefits: The impacts of inequitable access to public transit are thought to be large, yet spatial-temporal inequalities in transit provision remain little understood and of peripheral interest in transit planning. Clear and transparent tools will allow these considerations to be integrated into planning practice and public discourse. Codesigning the indicators and tools can empower policy makers to make bolder decisions with greater public support. Including equity indicators in transit performance assessments will help uncover inequalities, enabling planners and advocates to understand how disadvantaged populations in different locations and time periods are, or are not, served by public transportation. For example, visualizations showing the frequency of public transit service across areas of different income levels and at different times of day can help practitioners identify where and when low-income populations are underserved, highlighting where interventions are most needed. Performance metrics will be disaggregated by income, race, gender, and other relevant population characteristics, in addition to their traditional spatial and temporal dimensions.

Facilitation of knowledge transition: This project aims to develop EquiTransit for planners and community members to better identify who has the most and least amounts of different transportation services, but even more crucially, where those populations are located and when are the services needed. Using spatial and temporal visualizations, planners and other stakeholders will be better able to address issues in equity in transportation access. This would be particularly beneficial for transit agencies to better understand where service changes or additions should be made to increase transit equity.

Public Transportation and Equity: Traditional transit performance evaluations optimize service for an average user by focusing on efficiency, ridership, and reliability. These are system level, operational performance metrics that frame transit evaluation in a way that does not engage with social outcomes. Yet, transit is most heavily relied upon by individuals who have the fewest alternatives (i.e., captive users). Evidence from Canadian sources supports this concern. In 2021, 17.6% of racialized commuters in large urban centres took public transit to work, compared with 6.4% of non-racialized commuters, and 13.1% of immigrants relied on transit versus 4.7% of those born in Canada (27). By May 2025, roughly one in four Black (25.7%), Latin American (25.3%), and Filipino (24.4%) commuters used transit, and the average transit commute had reached 44.1 minutes, nearly twice that of a car commute, with most racialized groups facing systematically longer journeys (28). These patterns coincide with material deprivation. Allen & Farber (25) estimate that close to one million Canadians across the eight largest CMAs live in low-income households in areas of poor transit accessibility, placing them at risk of transport poverty. Women in lower-income households also undertake substantially more care-related travel that are poorly serviced by current network designs (30).

This makes the distribution of service to different people, at diverse places, and various times a question of justice (25). Distributive justice in transportation asks who benefits from the system and who carries its burdens, shifting the evaluative criterion from system performance to the accessibility opportunities delivered to different socioeconomic and demographic groups (26). This framework moves beyond horizontal equity, which treats all riders identically, to vertical equity, that is, actively directing better service, coverage, and frequency to the groups who rely on it most and face the greatest systemic barriers. Vertical equity represents a stricter idea of justice, as new benefits are unequally distributed in favor of under-privileged populations.

Existing planning tools rarely delve into these inequities. A recent analysis of Toronto's bus network shows that although two-thirds of TTC riders are equity-deserving, most live outside the agency's designated equity priority areas, leaving much of the population the framework is meant to serve largely invisible to it (20). Traditional performance metrics, and even most accessibility measures, also average across the day, masking systematic temporal under-service during evenings, nights, and weekends, when shift workers, who are disproportionately racialized, immigrant, and low-income, most depend on transit (17). EquiTransit will close these gaps with indicators and tools that bring distributive justice into transit planning across both space and time.

WORK PACKAGES

WP.1 Theoretically-grounded equity indicators: It is well established that the burdens and benefits of transportation are distributed unevenly across space and time, with some groups receiving greater benefits at the expense of others (27). To better understand how these benefits and burdens are distributed among a diverse population, a theoretical framework related to equity and justice is needed. In this proposal, we follow recent work of the PI with members of the SSHRC-funded project Mobilizing Justice team (29) that adapted the framework of Jaggar (30) to the needs of transportation concerns. According to this framework, the examination of equity, fairness, and justice is guided by six interrelated questions: *Why?*, *Who?*, *What?*, *Where?*, *When?*, and *How?*, or the 5WH framework.

Why? Justice is a broad moral and political concept that relates to: (1) the rights and entitlements that should be recognised and enforced; (2) the fairness of processes and procedures in decision-making and distribution (procedural justice); and (3) the appropriate distribution of benefits and burdens of social life in society. The last concept, also known as distributive justice, is the most studied form of justice in the transportation literature and aims to provide a fair redistribution to repair past harms (31). This is not a technical question, but a question of values, and therefore it is incumbent on the research team to involve stakeholders to avoid imposing perspectives that may be extraneous to the community the research is meant to benefit. CODESIGN (WP.2) is essential to this purpose.

The questions of *Who?*, *What?*, *Where?*, and *When?* in the 5WH framework all intersect with the spatial and temporal distribution of transit and the populations it serves. For this reason, they are better investigated from a geo-spatial perspective (WP.3). These questions are described next.

Who? Disadvantaged population groups, including those affected by low income, limited ability, and historical discrimination, have become a key focus of transportation research (31,32). This is due to the potential of historical discriminatory policies and other disadvantages to negatively impact their transport experiences. Additionally, these groups have often been excluded from the transportation planning process, resulting in their needs frequently being not recognized. Relevant groups include youth, adults, women, children, low-income individuals, zero-vehicle households, people with disabilities, single-parent households, refugees, and racialized populations (33–38).

What? In this research proposal, we argue that public transportation should be the focus of distributive justice. Public transportation is essential for promoting more sustainable cities due to its potential to reduce traffic congestion and carbon emissions and to improve air quality (39–41). In addition, public transportation increases people's accessibility, the main benefit of the transportation system (27), understood as the ease of reaching spatially distributed opportunities (42,43). In fact, public transportation provides access to opportunities for those who cannot - or prefer not to - drive, walk, or cycle. Access to opportunities and destinations is the most auspicious indicator for distributive justice (27), the most comprehensive indicator of the performance of land use and transport systems (44), and one of the main outcomes of spatial development (45) reduce the risk of social exclusion. Since transit and its output, accessibility, depend on space and time conditions, Scholars argue that this measure serves as a comprehensive and transparent indicator of transportation supply, with the additional advantage of being replicable across different jurisdictions and periods (45). Accessibility is a required condition for increasing people's freedom of choice and promoting equality of opportunities (45). An adequate level of access to opportunities, which can be offered by well-planned public transportation system, can increase the chances of attending a doctor's appointment, watching a movie at a cinema, engaging in a sporting activity in a park, or finding and keeping a job.

Where? By their very nature, interventions in the transportation sector are spatial interventions, and, due to the structuring effect of space (46), they inevitably generate inequality by benefitting those who live in or have access to the areas served by transportation infrastructure and services, while at the same time potentially harming others, either directly or indirectly (29). It is not uncommon for transportation policies to provide a sufficient or more than sufficient level of accessibility to the majority of people who own cars (47), while failing to ensure a basic level of accessibility for those who do not have access to a car, cannot use one, or do not wish to use one. To address this imbalance, improvements are needed in public transportation networks that provide users with a level of freedom comparable to that enjoyed by car users (48). The spatial dimension of analyses of transportation equity is often based on the concept of spatial equity, according to which cities should seek to distribute the benefits of transportation systems equally across space (31). Although transportation equity requires spatially targeted policies and interventions (34), rather than broad approaches that encompass all of society, our

perspective views equity primarily as an issue concerning people. Consequently, while we intend to identify areas with insufficient public transportation service and accessibility, these areas will be identified through measures and analyses focused on equity-deserving populations and facing insufficient access to public transportation, rather than solely on areas with low service. From this perspective, areas with a higher concentration of disadvantaged populations should take priority over wealthier areas, as disadvantaged groups tend to rely more on public transportation (28). Our work will be based on the City of Hamilton and its transit provider, Hamilton Street Railway (HSR).

When? To ensure justice in transportation, we evaluate the temporal aspects of transportation interventions. For instance, we investigate when a specific intervention takes place, the historical policy context in which it occurs, and for how long its benefits and burdens can be attributed to that intervention. In addition, the temporal dimension is central to our analysis because we are interested in investigating how equity-deserving populations may receive insufficient public transportation service, identifying not only the areas but also the times of day when this insufficiency is most pronounced.

The last question in our conceptual framework relates to How can measures be implemented to reduce inequities. The last work package of this project (WP.4) relates to the development of tools for equity-informed planning and evaluation. This part of the project intersects with WP.2 (CODESIGN) since these tools must implement concepts that are perceived as legitimate by all stakeholders.

How? Justice theories typically address sufficientarian and egalitarian moral concerns regarding how benefits and burdens should be distributed. While these two concepts are frequently applied in transportation equity research, they are often not explicitly articulated (22,49). The first approach, the sufficientarian perspective, focuses on why some groups or areas lack sufficient access to those goods and services. This approach builds on the idea of a ‘sufficient threshold’, that is, minimum levels of transport goods, services, and accessibility to essential activities that should be guaranteed to everyone to meet basic needs and lead to a meaningful life. In this idea, absolute priority should be given to improving the well-being of those whose well-being level falls below the sufficient threshold. Once that indicators have been selected and agreed upon by participants in the codesign part of the project, they will be implemented into equity-informed evaluation tools, as described in WP.4.

More generally, transportation analyses in practice often on standards to determine whether a situation is fair (29). Once operationalized, an equity standard defines the level of fairness to be adopted and the acceptable level of inequality, usually combining a concept of fairness with threshold criteria, thereby reflecting both vertical and horizontal approaches to equity. Vertical equity is typically operationalized through opportunity-based standards defined for specific population groups (e.g., minimum access to opportunities within a 10-minute bus ride for older adults), whereas horizontal equity is generally operationalized through uniform travel barrier thresholds applied to all potential users (e.g., a maximum acceptable distance to a public transit stop). Regarding the definition of equity standards, our approach is guided by the principles of sufficiency and egalitarianism. This means that the minimum level of accessibility to public transportation should be defined according to the needs of the most equity-deserving population, thereby reducing inequalities in opportunities.

WP.2 Codesign: In the 5WH framework, the question of WHY? constitutes the cornerstone of our approach to analyzing equity. This question, however, is not answerable in a strictly technical fashion. Data can help to support it, but to produce truly actionable knowledge public transit users and practitioners must co-create the equity indicators developed to assess the performance of public transport systems in Canadian cities. For this reason, CODESIGN is an essential component of the project, and will run throughout the life-course of the project, from the development of indicators, the choice of standards, and the testing of open-source tools.

Codesign is a form of community-based participatory action research; it is a collaborative approach that brings together researchers and stakeholders to address the needs of the community, enable social system change, and increase the impact of research. Stakeholders are placed at the centre of the design process to effectively solve problems and improve human experiences (50–52). In this project, the stakeholders will include the research team, public transit providers, and users.

Though the tools for this overarching proposed research project will be developed for Canada's 14 largest CMAs, the Codesign process will focus on the City of Hamilton, where McMaster is located and where the strongest community-based research already partnerships exist. Research collaborators working at the HSR will be involved as the public transit providers. Public transit users will be selected from Hamilton's 'code red' neighbourhoods; areas that experience the highest rates of poverty, unemployment, and chronic illness in the city (53).

The research will follow the 5-step human centered Codesign approach (54). Empathy is the first step. We will begin by understanding service users' experiences with public transport systems as well as their understandings of the standards the currently exist. This will be done through a series of Codesign focus groups. The second and third steps, to Define the key issues, and to Ideate, i.e., to brainstorm possible solutions, will take place in a second round of focus groups. This time, a series of separate Codesign focus groups will take place with users and with transit providers. PhD student 1 will lead these three sets of focus groups and present the results to the larger research team. The larger team will then complete step 4, which is to Prototype a variety of tangible solutions based on the focus group discussions so far. The final step is to Test and re-design the solution as needed. A final round of focus groups with users and providers in which they will present the prototyped solutions for feedback. These results will be presented to the research team again, with the possibility of another round of re-design and testing. The final outputs should include new tools that are not only theoretically and empirically sound, but that also respond to the needs of transit users and providers.

All Codesign focus group sessions will be supported by McMaster university's equity-based Co creation hub (eqCC). This hub unites researchers, students, service providers, and users to engage educate and innovate for systemic transformation (55). All focus groups will take place at McMaster University, and all participants will be compensated for their time with a \$50 gift cards per focus group they participate in. They will also be reimbursed for transportation (either reimbursed a bus fare or parking will be paid for).

WP.3 Geospatial data analytics: EquiTransit will be built entirely from open data and open-source software, ensuring transparency and reproducibility across jurisdictions. For each Census Metropolitan Area, we assemble four open inputs: sociodemographic data from the 2026 Canadian Census (via the {cancensus} interface to CensusMapper from von Bergman et al. (56), General Transit Feed Specification (GTFS) schedules published by transit agencies, OpenStreetMap road and pedestrian networks, and the Global Human Settlement Layer built-up surface. To enable consistent comparison, these inputs are harmonised onto a fine hexagonal grid; dasymetric interpolation, restricted to residential cells, apports census population and attributes to each hexagon. Using the {r5r} routing engine (57) we then compute different measures of transportation accessibility. This includes walking travel times between populated hexagons and transit stops to estimate the population effectively served by each stop. Larger network accessibility analyses will also be conducted to capture how well the transit system enables individuals to reach key destination types, such as places of work or daily necessities, around each study area within a reasonable travel time.

Critically, the travel time and accessibility analyses will be conducted using ten-minute transit service intervals, which is done to capture how level of service (e.g. service frequency) varies through the day and expose temporal gaps that daily averages conceal. GTFS Realtime data, which captures actual

transit vehicle movements rather than static schedules, offers a particularly interesting avenue to better understand how service levels and accessibilities are provided. Building on initial schedule-based analyses, we will expand our lens to incorporate realtime data over sampled time windows as the project progresses, making use of new computational tools like the {gtfsrealtime} (58) package for R. From this, the transit accessibilities are disaggregated by income, race, gender, and immigration status, revealing inequities across both space and time. The complete workflow will be packaged as a reproducible analytical pipeline and an interactive dashboard, initially covering Canada's fourteen largest CMAs, with parameters, indicators, and presentation refined through stakeholder consultation.

WP.4 Equity-informed transit planning and evaluation: Transit planning is often planned on legacy routes and updated as population and urban geography expand. Then the service is packaged into a global universal standard: General Transit Feed Specification (GTFS). While the planning process itself is mathematically challenging, it is fundamentally developed under a cost-reduction framework, maximizing the utilization of resources, and balancing operation with political decision and budget constraints. The well-established four-step transit design model (e.g., routes, frequencies, fleet, and crew assignment) does not include socio-economic and demographic factors that capture the equity dimension users. We will develop two interventions enabling a novel implementation of equity transit planning.

For existing network, a GTFS-based equity analytical model will be conceived, developed and deployed. The model will utilize equity indicator, geospatial data, and findings from the CODESIGN, and integrate them with GTFS files. Furthermore, Equity KPIs will be embedded in a GTFS-analytical model, enabling transit agencies, policymakers, and practitioners to evaluate the equity gaps in existing design. For new service design, the equity indicators will be integrated in the four-step transit model, offering a novel equity-based transit planning framework transferable to any context. Equity-indicators will be integrated in all four steps of the planning process. For example, frequency estimation is based on passenger flow/demand, wait-time, and fleet size (e.g., available resources). Although passenger flow/demand is an indirect indicator of equity, it remains ineffective of differentiating service provision for equity-deserving groups compared to other users.

Table T1. Project timeline with Work Packages and Tasks.

Work Packages	Tasks	Year 1	Year 2	Year 3	Year 4	Year 5
<i>WP.1 Theoretically-grounded equity indicators</i>	<i>Overall direction and transversal communication between work packages (Why? What? Who? Where? When? How?)</i>	Antonio Pérez & Juanita De Barros				
		PD 1				PD 2
<i>WP.2 Codesign</i>	<i>Equity indicators (Why?)</i>	Dr. Léa Ravensbergen				
	<i>EquiTransit tools (How?)</i>	PHD 1			PHD 1	PD 2
<i>WP.3 Geospatial data analytics</i>	<i>Equity indicators (What? Who? Where? When?)</i>	Dr. Christopher Higgins				
	<i>EquiTransit tools (How?)</i>	PD 1	PHD 2			PHD2
<i>WP.4 Equity-informed transit planning and evaluation</i>	<i>Development of base tools</i>	Dr. Moataz Mohamed				
	<i>Integration of equity indicators (How?)</i>	M 1			M 2	

Personnel (Key). PD1, 2: Post-Doctoral Fellows; PHD 1, 2: Doctoral Students; M1, 2: Master's Students